

Minimum Beta Notes

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Minimum β for Convexity of

$$h_\beta(x) = f(x) + \beta g(x)$$

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1 Problem Statement

Let $f, g : A \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ be twice differentiable convex functions. We aim to determine the minimum $\beta \in \mathbb{R}$ such that the function

$$h_\beta(x) := f(x) + \beta g(x)$$

remains convex on the domain A .

2 Theoretical Background

A twice differentiable function $h : A \rightarrow \mathbb{R}$ is convex on an open convex set $A \subset \mathbb{R}^d$ if and only if its Hessian $\nabla^2 h(x)$ is positive semidefinite (PSD) for all $x \in A$.

Since both f and g are convex and smooth, $\nabla^2 f(x) \succeq 0$ and $\nabla^2 g(x) \succeq 0$ for all $x \in A$. We seek the smallest scalar $\beta \geq 0$ such that:

$$\nabla^2 h_\beta(x) = \nabla^2 f(x) + \beta \nabla^2 g(x) \succeq 0 \quad \forall x \in A.$$

Assumptions and Smoothness

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a twice continuously differentiable function. This means:

- $f \in C^2$ (all partial derivatives up to second order exist and are continuous).
- The gradient is defined as:

$$\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle$$

- The Hessian matrix is:

$$\nabla^2 f = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{i,j}$$

which is symmetric because mixed partials are equal due to Clairaut's theorem.

Clairaut's Theorem (Equality of Mixed Partial)

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, if all second partial derivatives exist and are continuous near a point $\vec{x}$, then the mixed partials satisfy:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i} \quad \text{at that point.}$$

Critical Points

To find critical points, solve:

$$\nabla f(\vec{p}) = \vec{0}$$

This means all partial derivatives vanish at $\vec{p}$. These are the candidate points for local minima, maxima, or saddle points.

Let $S := \nabla^2 f(\vec{p})$ be the Hessian evaluated at a critical point $\vec{p}$.

We analyze the matrix S by looking at its eigenvalues.

Positive Semidefinite Condition

$$S \succeq 0 \iff x^\top S x \geq 0 \quad \forall x \in \mathbb{R}^n \iff \lambda_i \geq 0 \quad \text{for all eigenvalues } \lambda_i \text{ of } S$$

Spectral Theorem

Since S is symmetric, it can be diagonalized:

$$S = Q D Q^\top$$

where:

- $Q \in \mathbb{R}^{n \times n}$ is an orthogonal matrix ($Q^\top = Q^{-1}$),
- $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a diagonal matrix of eigenvalues.

Signature and Classification

Let the signature of S refer to the sign pattern of the eigenvalues:

- **All** $\lambda_i > 0$: S is positive definite $\rightarrow$ strict local **minimum**
- **All** $\lambda_i < 0$: S is negative definite $\rightarrow$ strict local **maximum**
- **Mixed signs**: S is indefinite $\rightarrow$ **saddle point**
- **All** $\lambda_i \geq 0$: S is positive semidefinite $\rightarrow$ test is **inconclusive**
- **All** $\lambda_i \leq 0$: S is negative semidefinite $\rightarrow$ test is **inconclusive**

Convexity Remark

Note: The second derivative test is **local**, and if the Hessian is only semidefinite at a critical point, the test is inconclusive.

However, if f is globally convex — that is, $\nabla^2 f(x) \succeq 0$ for all $x \in \mathbb{R}^n$ — then **any** critical point is guaranteed to be a global minimum.

Similarly, if f is concave (i.e., $\nabla^2 f(x) \preceq 0$ everywhere), then any critical point is a global maximum.

3 Reformulated Optimization Problem

We solve:

$$\min \beta \quad \text{such that} \quad \nabla^2 f(x) + \beta \nabla^2 g(x) \succeq 0 \quad \forall x \in A.$$

This is a semi-infinite optimization problem with infinitely many PSD constraints.

4 Numerical Strategy

Discretize A with points $x_1, \dots, x_n \in A$. For each x_i , compute the smallest β such that:

$$\nabla^2 f(x_i) + \beta \nabla^2 g(x_i) \succeq 0.$$

Let $\lambda_{\min}^{(i)}(\beta)$ be the smallest eigenvalue of $\nabla^2 f(x_i) + \beta \nabla^2 g(x_i)$. The condition becomes:

$$\lambda_{\min}^{(i)}(\beta) \geq 0.$$

If $\nabla^2 g(x_i) \succ 0$, $\lambda_{\min}^{(i)}(\beta)$ is increasing in β . Solve for minimal β_i where $\lambda_{\min}^{(i)}(\beta_i) = 0$. The overall minimum β is:

$$\beta^* = \max_{1 \leq i \leq n} \beta_i.$$

5 Final Answer

The minimal β is:

$$\beta^* = \sup_{x \in A} \inf \{ \beta \in \mathbb{R} : \nabla^2 f(x) + \beta \nabla^2 g(x) \succeq 0 \}$$

Approximate via dense sampling or convex programming with matrix inequalities.